MIL STD-1437
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MILITARY STANDARD

ORGANIC CHEMICAL COMPOUNDS, SOLID, TECHNICAL GRADE

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FSC 6810

DEPARTMENT OF DEFENSE
Washington, D.C. 20301

Organic Chemical Compounds, Solid, Technical Grade

MIL-STD-1437

1. This Military Standard is mandatory for use by all departments and agencies of the Department of Defense, to assure that selection of new items is limited to essential items, for which no comparable standard item exists. This document is not intended to restrict any service in selecting new items required to support state-of-the-art changes.

2. Recommended corrections, additions, or deletions should be addressed to Commanding Officer, Edgewood Arsenal, ATTN: SMUEA-TSE-SM, Edgewood Arsenal, Maryland, 21010.

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FOREWORD

This is the first book format generated on Organic Chemical Compounds, Solid, Technical Grade. This document is mandatory for use by all departments and agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This is not a procurement document. This document is not intended to restrict any service in selecting new items required to support state-of-the-art changes.

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1. SCOPE

1.1 Coverage. This standard is a presentation of nomenclature, symbols, physical and chemical properties and requirements, military and typical commercial uses, directions for use, packaging data, labeling, general safety precautions, storage information, and shelf life of all military standard technical grade, solid, organic chemical compounds. This standard does not necessarily include all classifications of the items represented by the title or those which are commercially available. Many solid, organic chemical compounds which may seem applicable to this standard are not included because they are either reagent grade or are included under other military standards covering technical grade chemicals; aldehydes, ketones and peroxides; esters and metal organics; organic acids; miscellaneous alcohols and amines; or liquid organic compounds. It does contain items preferred for use in the selection of technical grade, solid organic chemical compounds for application by the Department of Defense. This standard covers the following seventeen items.

NAME	NO. OF ITEMS
ACACIA, TECHNICAL	4
ALBUMEN, EGG, TECHNICAL	1
CORNSTARCH, TECHNICAL	1
p-DICHLOROBENZENE	2
GELATIN, TECHNICAL	1
GUM GHATTI, TECHNICAL	1
METHYL CELLULOSE, TECHNICAL	1
NAPHTHALENE, TECHNICAL	2
POLYETHYLENE GLYCOL, TECHNICAL	1
ROSIN, GUM	2
UREA, TECHNICAL	1

1.2 Application. Items listed herein accommodate essential requirements of the military and defense agencies, and will effect continued economies in all logistics functions where properly employed in new applications.

2. REFERENCED DOCUMENTS

The issues of the following documents in effect on the date of invitations for bid form a part of this standard to the extent specified herein.

Federal Specifications

C-A-450	Albumen, Egg, Technical, Scale Form
O-P-99	Paradichlorobenzene, Technical (Dichlorobenzene, Para)
R-N-91	Naphthalene, Technical
JJJ-A-20	Acacia, Technical (Gum Arabic)

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LLL-R-626	Rosin, Gum; Rosin, Wood; and Rosin Tall Oil
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals
PPP-C-301	Chemicals, Dry and Paste, Packaging and Packing of

Military Specifications

MIL-C-21963	Cornstarch, Technical (Boiler Water Filtration)
MIL-U-10866	Urea, Technical

Military Standards

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-12	Abbreviations for use on Drawings, Specifications, Standards and in Technical Documents

3. GLOSSARY

3.1 Definitions

Amorphous - An amorphous substance is one that has no definite form or is not crystallized.

Colloid - The stage between solution and suspension of a solid in a liquid with no apparent settling. In general, particle diameters range from about 0.005 microns to about 0.2 microns in such colloidal solutions (suspensions).

Decomposition - The chemical separation of a substance into two or more simpler substances, which differ from each other and from the original substance.

Explosive limits - When combustible vapor is mixed with air in the proper proportions, ignition will produce an explosion. This proper proportion is called the explosive range. The explosive range includes all concentrations of a mixture of flammable vapor or gas in air, in which a flash will occur or a flame will travel if the mixture is ignited. The lowest percentage at which this occurs is the lower explosive limit; and the highest percentage, the upper explosive limit. Explosive limits are expressed in percent by volume of vapor in air and, unless otherwise specified, under normal conditions of temperature and pressure.

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Flash point - The temperature to which a substance must be heated under specific conditions to give off sufficient vapor to form a mixture with air that can be ignited momentarily by a specified flame.

Formula weight - The sum of the atomic weights of all the atoms appearing in a chemical formula. In this standard, it is computed according to international atomic weight values of 1961.

Hazardous substance - Any substance or mixture of substances which is (1) toxic; (2) corrosive; (3) an irritant; (4) a strong sensitizer; (5) flammable, or which (6) generates pressure through decomposition, heat, or other means, if such substance or mixture of substances may cause substantial personal injury or substantial illness during or as a direct result of any customary or reasonably anticipated handling or use.

Melting point - The temperature at which solid and liquid forms of a substance are in equilibrium, and transition from the solid to the liquid occurs.

Saponification number. The number of milligrams of potassium hydroxide required to saponify one gram of a sample of an ester or mixture.

Solubility - The weight of a substance which will dissolve in a specific volume of solvent at a specified temperature to produce a saturated solution. In this standard, the solubility will be expressed as the weight of solute in grams dissolved in 100 ml of the specific solvent.

Specific gravity - The ratio of the mass of a body to the mass of an equal volume of water at 4°C or other specified temperature. In this standard, the first temperature indicates the temperature of the substance and the second indicates the temperature of the water to which it is referred. If there is no mention of the temperatures, (20/4°C) is understood.

Technical grade - Denotes a quality of chemicals which are generaly used for industrial, solvents, and manufacturing applications. Generally, specific processes are not employed by the manufacturer to limit all the impurities, aside from the normal precautions which are taken in the manufacturing process. A technical chemical may be specially processed to reduce specific impurities so as to suit the chemical to a given industrial application. In such cases, the identification of the items must be further expanded to indicate the specific impurities limitation.

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Vapor pressure - The pressure exerted when a solid or liquid is in equilibrium with its own vapor. The vapor pressure is a function of the substance and of the temperature.

Volatile substance - A substance which evaporates rapidly due to its high vapor pressure.

3.2 Abbreviations. The same abbreviation is used for all tenses, the possessive case, and the singular and plural forms of a given word.

Be' - Baume'

C - Celsius (Centigrade)

F - Fahrenheit

FW - Formula Weight

g - gram

max - maximum

mg - milligram

MIL-STD - Military Standard

min - minimum

ml - milliliter

mm - millimeter

pct - percent

pH - Hydrogen ion concentration

wt - weight

4. GENERAL REQUIREMENTS

4.1 Chemical and physical requirements. When military or federal specifications establish requirements for degrees of purity of the chemicals listed herein, those chemicals and physical properties affected are tabulated as

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requirements. These requirements are subject to change as specifications are revised. When no specific limitations are imposed or no specification exists, the chemical and physical properties are tabulated as constants or typical properties.

4.2 Nomenclature. All chemicals in this standard conform to the military definition of technical grade as set forth under section 3.1. The Department of Defense item names, as used throughout this standard, are in capital letters. Other names that are sometimes used commercially are in small letters immediately beneath.

4.3 Packaging data and labeling. All organic chemicals included in this standard shall be packaged in accordance with Federal Specification PPP-C-301 and with all applicable documents mentioned in this document. The precautionary labeling prescribed in this standard is not intended to replace or substitute for precautionary labeling required by federal or state laws or regulations. When labels required by statute contain essentially the same information, the label prescribed by this standard is not required in addition thereto.

4.4 Safety. Only general safety precautions, where applicable, are outlined in this standard. For more specific information the appropriate medical or safety authorities must be consulted.

4.5 Shelf life. Factors such as moisture, temperature, type and and condition of container, and exposure to sunlight and the atmosphere cause variations in shelf life. Ideal storage conditions are outlined for each item. The term "cool" denotes temperatures from freezing up to 110°F but not consistently over 100°F when stored out of direct sunlight. The term "dry" is usually used to denote an area where condensation does not come in contact with the packages or contents (for example, storing on pallets away from walls in an enclosure or building). Periodic examinations of the material should be made more frequently when storage conditions vary from the ideal. For applications where quality may be critical, each compound should be analyzed prior to use. Shelf life is dated from the date of manufacture. Unless otherwise specified, it is preferable that not more than one year shall have elapsed from date of manufacture to date of purchase.

4.6 Solubility data. Solubility data is given for only the most common solvents.

4.7 Temperature. If the temperature at which a property was determined is not specified, it is understood to be room temperature (20° to 25°C. or 68° to 77°F).

4.8 Substitutability and interchangeability. None of the chemicals in this standard are completely interchangeable with other chemicals in any given circumstance. One chemical may provide the same end result as another, but procedures, techniques, and other factors would have to be altered to do so. Time, economy, and the discretion of the user would govern such circumstances.

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5. DETAIL REQUIREMENTS

5.1 Name. ACACIA, TECHNICAL
Gum Arabic; Senegal; Kordofan; Morocco;
Cape; Aden; Suakin; White Senaar; Wattle Gum.

5.1.1 Specifications. JJJ-A-20; Acacia, Technical (Gum Arabic)

5.1.2 Technical description. Acacia is the dried exudate from the stems and branches of Acacia Senegol (Linne) Willdenow or of other related African species of Acacia (Family Leguminosae). A carbohydrate polymer, complex and highly branched; it is composed essentially of the calcium salt of arabic acid and on hydrolysis, yields arabinose, galactose, aldobionic and galacturonic acids. It varies in form of lump, thin flakes, powder, granules or angular fragments of white to yellowish white color, and it is almost odorless. Acacia is completely soluble in hot or cold water, yielding a uniform viscous solution of mucilage which is acid to litmus. It is insoluble in alcohol. The specific gravity is 1.35 to 1.49, but samples dried at 100°C are heavier. Moisture content usually varies from 13 to 15%.

In addition to the requirements listed in Table I, the Acacia shall be free from tannin-bearing gums, starch and dextrin, and inorganic acidity. It shall not reduce Fehlings solution and shall pass the test for identification as outlined in Federal Specification JJJ-A-20.

TABLE I - Chemical and physical requirements of acacia, technical

Property	Requirement (max) pct by wt
Acid-insoluble salt	0.5
Ash, total	4.0
Insoluble residue	1.0
Moisture	15.0
Organic acidity, as acetic acid	0.4

5.1.3 Use data. Acacia, Technical is intended for military use as a colloidal stabilizer in photographic emulsions and for lithographic work. Typical commercial applications include use in adhesives; inks; textile printing; in general, as a thickening agent and colloidal stabilizer.

5.1.4 Packaging data and labeling. Acacia, technical is packaged for military use in one and 20 pound unit quantity cans for lump form, in one pound unit quantity cans for granular form and one pound unit quantity bottles for powder form.

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5.1.5 Storage data. Acacia, technical should be stored in closed containers in a cool, dry place. When stored under these conditions, the shelf life is indefinite.

5.2 Name. ALBUMEN, EGG, TECHNICAL
Ovalbumin
Albumen

5.2.1 Specifications. C-A-450; Albumen, Egg, Technical, Scale Form.

5.2.2 Technical description. Egg albumen is the dried white of egg separated from the egg yolk. It is pale yellow, transparent, in amorphous lumps or scales, free from egg yolk, dyes or other denaturants and foreign matter; a simple naturally occurring protein soluble in water and coagulated by heat. Egg albumen will decompose in moist air with the formation of ammonia. Technical grade egg albumen intended for military use shall be free from starch, dextrin, glucose and reducing sugars. It shall have the appearance of shiny, pale yellow scales and have a characteristic egg odor, free from objectionable odor.

TABLE II. Chemical and physical requirements of albumen, egg, technical

Property	Requirement
Non-volatile matter (pct by wt, min)	85
pH of aqueous solution (20%)	5.0 to 6.5
Specific gravity of 20% aqueous solution (25°C)	1.037 (5.2°Be')
Water-insoluble matter (pct by wt, max)	3

5.2.3 Use data. Albumen, egg, technical is intended for military use in the preparation of sensitizing solutions for photolithographic zinc, aluminum, plastic, glass or composition plates. Typical commercial applications include use as a clarifying agent. It has the property of coagulating when heated and carrying down coloring matter and other impurities in the process. It is also used in the manufacture of cements, adhesives, varnishes, ivory substitutes; in photography for manufacture of albumen paper and coating plates; sizing paper and printing fabrics; rendering certain dyes fast; stamping with gold and bronze powder and leather manufacture.

5.2.4 Packaging data and labeling. Albumen, egg, technical is packaged for military use in one pound unit quantity bottles.

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5.2.5 Storage data. Technical grade egg albumen should be stored in tightly closed containers in a cool, dry place. When stored under these conditions, the shelf life is indefinite.

5.3 Name. CORNSTARCH, TECHNICAL
Amylum

5.3.1 Specifications. MIL-C-21963; Cornstarch, Technical (Boiler Water Filtration)

5.3.2 Technical description. Cornstarch, technical is a white, amorphous, tasteless, fine powder prepared from clean, sound, corn (maize) that is essentially free from other grains, weed seeds and other foreign material. It has a specific gravity range of 1.499 to 1.513; does not melt but decomposes (burns) when heated. It is insoluble in cold water, alcohol or ether, however is soluble in hot water forming a jelly on cooling. Cornstarch, technical shall be made of such purity to meet requirements of Table III and shall be finely pulverized, white to cream in color, be free of lumps, caking, dirt and any foreign matter.

TABLE III. Chemical requirements of cornstarch, technical

Property	Requirement, pct, max
Ash*	0.15
Moisture	12.0
Nitrogen*	0.08

*Determined on a moisture-free basis.

5.3.3 Use data. Cornstarch, technical is intended for military use in the treatment of sea water feed to marine distillers aboard ship. Typical commercial applications include use in manufacture of adhesives; sizing and finishing textiles; sizing paper; book binding; and colloidal preparations.

5.3.4 Packaging data and labeling. Cornstarch, technical is packaged for military use in sealed, siftproof, five pound unit quantity boxes. Each container shall bear the following label:

CAUTION! COMBUSTIBLE SOLID
Keep away from heat, sparks, and open flame.
Keep container closed.

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5.3.5 Safety precautions. When exposed to flame, starch dust may ignite or explode and can react with oxidizing materials. Care in storage and handling must be exercised.

5.3.6 Storage data. Cornstarch, technical should be stored in a cool, dry place, away from definite fire hazards or oxidizing materials. When stored under these conditions, the shelf life is indefinite.

5.4 Name. p-DICHLOROBENZENE, TECHNICAL $C_6H_4Cl_2$ FW 147.00
Paradichlorobenzene
1,4-Dichlorobenzene
PDB
(HAZARDOUS)

5.4.1 Specifications. O-P-99; Paradichlorobenzene, Technical (Dichlorobenzene, Para)

5.4.2 Technical description. p-Dichlorobenzene, technical is a white crystalline, volatile material of characteristic penetrating odor, and freely soluble in alcohol, benzene, chloroform, ether, and carbon disulfide, but insoluble in water. It has a specific gravity of 1.458; boiling point of 173.7°C; melting point of 53°C and a flash point of 150°F. It is derived by the further chlorination of monochlorobenzene under proper conditions.

TABLE IV. Chemical and physical requirements of p-Dichlorobenzene, technical

Property	Requirement
Crystallizing point (°C)	52.5 to 53.5
Particle size:	
Passing 3/8 in. sieve (pct, min)	100
Passing No. 20 sieve (pct, max)	5

In addition to requirements listed in Table IV, the crystals, when melted, shall be essentially free from turbidity and suspended matter.

TABLE V. Physical constants of p-Dichlorobenzene

Property	Constant
Boiling point (°C)	174
Flash point (closed cup, °F)	150
Melting point (°C)	53
Specific gravity (21/4°C)	1.458
Vapor density (air = 1.00)	5.07

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5.4.3 Use data. p-Dichlorobenzene, technical is intended for military use as an insecticide, primarily as a moth preventative. Typical commercial applications include use in insecticides; germicides; deodorants; dyes; agricultural soil and crop fumigating products.

5.4.4 Packaging data and labeling. p-Dichlorobenzene, technical is packaged for military use in one pound unit quantity cans and 100 pound unit quantity drums. Each container shall bear the following label:

p-DICHLOROBENZENE
CAUTION! DUST AND VAPOR IRRITATING

Keep away from heat and open flame.
Store away from oxidizing materials.
Use with adequate ventilation.
Avoid prolonged breathing of dust or vapor.
Do not get in eyes. In case of contact,
immediately flush with plenty of
water for at least 15 minutes and
get medical attention.

5.4.5 Safety precautions. p-Dichlorobenzene, technical, is not compatible with naphthalene. Care should be taken not to use these two items interchangeably. p-Dichlorobenzene can react vigorously with oxidizing materials and when heated to decomposition releases vapors which are very painful to the eyes and nose. Under certain conditions of use, protective equipment may be required. In order to determine personal protective measures and environmental controls, the appropriate medical or safety authorities must be consulted.

5.4.6 Storage data. p-Dichlorobenzene, technical should be plainly labeled and stored in tightly closed containers in a cool ventilated area, away from oxidizing materials, open flames and areas of acute fire hazard. When stored under proper conditions, the shelf life is indefinite.

5.5 Name. GELATIN, TECHNICAL

5.5.1 Specifications. None

5.5.2 Technical description. Gelatin, technical is a protein obtained from collagen by boiling skin, ligaments, tendons, bones, principally from beef and calves, with water. It is a colorless or slightly yellow, practically odorless and tasteless powder that swells up and absorbs five to ten times its weight of cold water. It is soluble in hot water, glycerol and acetic acid, but insoluble in alcohol, chloroform and other organic solvents.

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5.5.3 Use data. Gelatin, technical is intended for military use in photography, lithography and adhesives. Typical commercial applications include use in photography and photolithography and as a protective colloid in electrometallurgical and other processes.

5.5.4 Packaging data and labeling. Gelatin, technical is packaged for military use in one kilogram unit quantity bottles.

5.5.5 Storage data. Gelatin, technical should be stored in tightly closed containers in a cool, dry place. When stored under these conditions, the shelf life is indefinite.

5.6 Name. GUM GHATTI, TECHNICAL
Ghatti Gum
Indian Gum
British Indian Gum

5.6.1 Specifications. None.

5.6.2 Technical description. Gum Ghatti, technical is somewhat similar to gum arabic (Acacia). It is the gummy exudate from the stem of the Anogeissus latifolia, Dhaua and related trees of India and Ceylon. It is in the form of colorless to pale yellow tears, rounded or vermiform. A tasteless and odorless substance that is partially soluble in water. It can be solubilized by autoclaving. Solubilized gum ghatti in solution, forms a very viscous mucilage, more viscous but less adhesive then acacia.

5.6.3 Use data. Gum Ghatti is intended for military use as a colloidal stabilizer in photographic emulsions and lithographic work. Typical commercial applications include use in adhesives; inks; textile printing; in general as a thickening agent and colloidal stabilizer. Gum ghatti can replace acacia in many cases.

5.6.4 Packaging data and labeling. Gum Ghatti, technical is packaged for military use in one pound unit quantity bottles.

5.6.5 Storage data. Gum Ghatti, technical should be stored in closed containers in a cool, dry place. When stored under these conditions the shelf life is indefinite.

5.7 Name. METHYL CELLULOSE, TECHNICAL
Cellulose Methyl Ether

5.7.1 Specifications. None.

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5.7.2 Technical description. A methyl ether of cellulose. It is a grayish white fibrous powder formed by reaction of methanol with cellulose in the presence of dehydrating agents. The proportions of the reacting materials are varied to control the properties of the product, such as water solubility and viscosity of the water solutions. This compound swells in water and produces a clear to opalescent, viscous, colloidal solution which is neutral to litmus. It is insoluble in hot water, alcohol, ether and chloroform but soluble in glacial acetic acid, ethyl acetate, ethylene dichloride, benzene and carbon tetrachloride. It is not affected by oils and greases, is stable to light and to 300°C. It is flammable when ignited.

5.7.3 Use data. Methyl Cellulose, technical is intended for military use as a dispersing agent, emulsifier and thickener. Typical commercial applications include use in fabric finishing "crease proofing", as an adhesive and as a sizing agent.

5.7.4 Packaging data and labeling. Methyl Cellulose, technical is packaged for military use in one pound unit quantity bottles. Each container shall bear the following label:

CAUTION! COMBUSTIBLE SOLID

Keep away from open flame.
Keep container closed.

5.7.5 Safety precautions. Methyl Cellulose, technical is flammable and due care should be exercised to keep it away from open flames.

5.7.6 Storage data. Methyl Cellulose, technical should be stored in closed containers in a cool, dry area away from acute fire hazards and open flames. When stored under proper conditions, the shelf life is indefinite.

5.8 Name. NAPHTHALENE, TECHNICAL $C_{10}H_8$ FW 128.18
Tar camphor
Naphthalin
(HAZARDOUS)

5.8.1 Specifications. R-N-91; Naphthalene, Technical

5.8.2 Technical description. Naphthalene, technical is a white, crystalline, volatile solid, derived from the coal-tar oils boiling between 170 and 230°C by crystallization and distillation or from petroleum fractions after various catalytic processing operations. It has the characteristic odor of moth balls and is insoluble in water, moderately soluble in absolute alcohol and very soluble in benzene. For military use, naphthalene shall solidify at a minimum temperature of 79°C; shall have no residue on ignition; shall be insoluble in water and soluble in boiling ethanol. It shall be supplied in ball or flake form.

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TABLE VI. - Physical constants of napthalene

Property	Constant
Boiling point (760 mm pressure, °C)	217.9
Explosive limits (pct by volume in air)	0.88 to 5.9
Flash point (closed cup, °F)	176
Melting point (°C)	80.2
Specific gravity (20/4°C)	1.145
Vapor density (air = 1.00)	4.42

5.8.3 Use data. Naphthalene, technical is intended for military use as an insecticide, primarily as a moth preventative. Typical commercial applications include use in the manufacture of dyes; moth repellents; fungicides; explosives; cutting fluids; lubricants; synthetic resins; synthetic tannings, preservatives and emulsion breakers.

5.8.4 Packaging data and labeling. Naphthalene, technical is packaged for military use in one pound unit quantity cartons for flake form and in one pound unit quantity boxes for ball form. Each container shall bear the following label:

NAPHTHALENE
CAUTION! IRRITATING TO THE SKIN
FLAMMABLE

Keep away from heat and open flame.
Store away from oxidizing materials.
Use with adequate ventilation.
Avoid prolonged breathing of dust and vapor.
Avoid contact with skin or eyes. In case of contact, immediately flush with plenty of water for at least 15 minutes; for eyes get medical attention.

5.8.5 Safety precautions. Naphthalene, technical, forms explosive mixtures in air and reacts with oxidizing materials. It is irritating to the skin and is readily absorbed when inhaled. Naphthalene is not compatible with p-Dichlorobenzene, also used as a moth repellent. Care should be taken not to use these two items interchangeably.

5.8.6 Storage data. Naphthalene, technical should be plainly labeled and stored in closed containers in a cool, dry place away from all fire hazards and oxidizing materials. When stored under proper conditions, the shelf life is indefinite.

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5.9 Name. POLYETHYLENE GLYCOL, TECHNICAL (approx) FW 7500
(approx) $H(OCH_2CH_2)_{170}OH$

5.9.1 Specifications. None.

5.9.2 Technical description. Polyethylene glycol, technical required for military use is a white, odorless, wax-like solid. It has an average molecular weight of 7500 and is soluble in water, alcohol and other organic solvents. It is inert to many chemical agents and has a low vapor pressure. Material of this molecular weight would have a specific gravity of approximately 1.212 (25/25°C) and a flash point at approximately 515°F.

5.9.3 Use data. Polyethylene glycol, technical is intended for military use as a lubricant, a mold release agent and an anti-sticking compound. Typical commercial applications include use as a coupling agent in cutting oils; as lubricants in metal rolling, forming and drawing; as antifoam agents in ceramic glaze solutions, latex paints, methanol antifreezes, rubber lubricants and mold release agents; lubricants in brake fluid formulations and as floatation froth control agents.

5.9.4 Packaging data and labeling. Polyethylene glycol, technical is packaged for military use in 50 pound unit quantity bags.

5.9.5 Storage data. Polyethylene glycol, technical should be stored in a cool, dry place. When stored under proper conditions, the shelf life is indefinite.

5.10 Name. ROSIN, GUM
Rosin
Colophony

5.10.1 Specifications. LLL-R-626; Rosin, Gum; Rosin, Wood; and Rosin, Tall Oil.

5.10.2 Technical description. Gum rosin is the residue obtained after the distillation of turpentine oil from the exuded oleoresin collected from living pine trees, chiefly Pinus palustris and Pinus caribaea. It is translucent, yellow, amber or red colored, in large lumps or solid mass. It is insoluble in water; freely soluble in alcohol, benzene, ether, glacial acetic acid, oils, carbon disulfide and dilute solutions of fixed alkali hydroxides. Gum rosin has a specific gravity of 1.08 at 25°C and a melting point of 100 to 150°C. Resin acids of the abietic and pimaric types have the general formula $C_{19}H_{29}COOH$, and have a phenanthrene nucleus. They are unsaturated and hence reactive. An unsaponifiable portion of rosin (3-10%) contains hydrocarbons and high molecular weight alcohols. These constituents vary greatly according to the source of the rosin.

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TABLE VII. Chemical requirements of rosin, gum

Property	Requirement
Acid number (mg KOH/g of sample, min)	160
Saponification number (mg KOH/g of sample, min)	166
Toluene-insoluble matter (pct by wt, max)	0.05

Gum rosin for military use shall be in the form of a solid mass of the darker colored rosins (grades D and E of the Federal Naval Stores Act). It shall be clear and transparent, free from more than traces of visible particles of dirt or foreign matter suspended in the solid mass. The softening point by Ring and Ball method shall not be lower than 70°C.

5.10.3 Use data. Rosin, fum is intended for military use as a matrix in shrapnel and in paints, varnishes and driers. Typical commercial applications include use in the manufacture of linoleum; soldering compounds; core oils; insulating compounds; molding compounds and sealing waxes.

5.10.4 Packaging data and labeling. Rosin, gum is packaged for military use in ten pound unit quantity cans and in 500 pound unit quantity drums. Each container shall bear the following label:

WARNING! FLAMMABLE
MAY IRRITATE SKIN AND EYES

Keep container closed.
Keep away from heat, sparks, open flame and oxidizing materials.

5.10.5 Safety precautions. Rosin, gum is flammable and gives off acrid fumes. It has a tendency to heat spontaneously and can react with oxidizing materials.

5.10.6 Storage data. Rosin, gum should be stored in closed containers in a cool, well ventilated area, away from all fire hazards, open flames and oxidizing agents. When stored under these conditions, the shelf life is indefinite.

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5.11 Name. UREA, TECHNICAL $CO(NH_2)_2$ FW 60.06
Carbamide
Carbonyldiamide

5.11.1 Specifications. MIL-U-10866; Urea, Technical.

5.11.2 Technical description. Urea, technical is supplied in the form of small, free flowing pellets or beads. It is almost odorless, has a specific gravity of 1.335 and decomposes before boiling. Urea is soluble in water, alcohol and benzene, slightly soluble in ether and almost insoluble in chloroform. Urea occurs in urine and other body fluids.

TABLE VIII. Chemical and physical requirements of urea, technical

Property	Requirement
Ash (pct by wt, max)	0.1
Biuret content (pct by wt, max)	2.5
Melting point (°C)	129.0 to 133.0
Moisture (pct by wt, max)	0.50
Nitrogen (pct by wt, max)	46.0 to 46.7

5.11.3 Use data. Urea, technical is intended for military use to melt thin ice at temperatures above 15°F as well as prevention of ice accumulation during freezing rain upon aircraft runways and ramp areas. Typical commercial applications include use in the manufacture of fertilizers, animal feed, resins; stabilizers for explosives, and in adhesives.

5.11.4 Packaging data and labeling. Urea, technical is packaged for military use in 100 pound unit quantity bags.

5.11.5 Safety precautions. Urea can react with oxidizing materials and when heated emits fumes of cyanide. Proper storage and handling will minimize this potential hazard.

5.11.6 Storage data. Urea, technical should be stored in a cool, dry place away from all fire hazards, open flames and oxidizing agents. When stored under proper conditions, the shelf life is indefinite.

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Assignee activity - Defense General Supply Agency

Custodians: Army - MU
Navy - YD
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